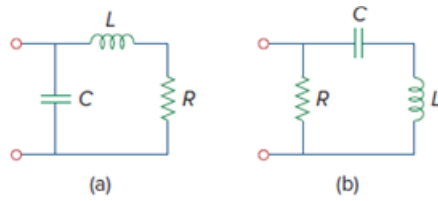


Problem

Calculate the resonant frequency of each of the circuits in Fig. 14.82.

Figure 14.82:



Step-by-step solution

Step 1 of 7

(a)

Consider the following circuit:

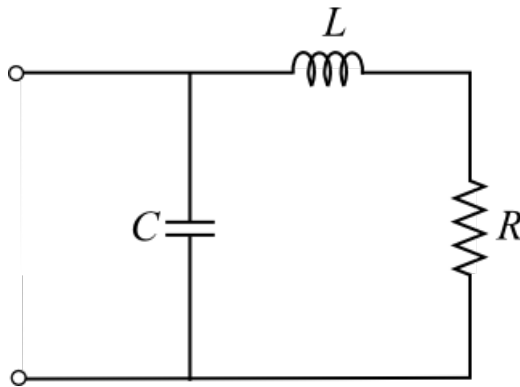


Figure 1

Step 2 of 7

The frequency domain equivalent circuit is as shown below:

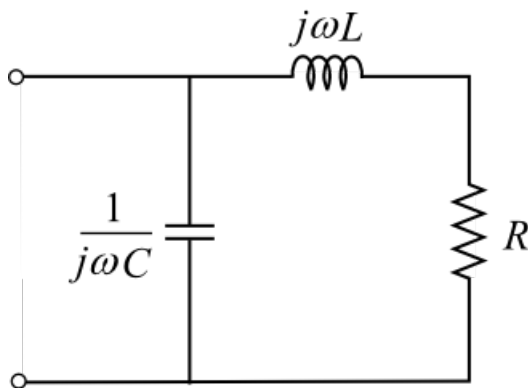


Figure 2

Step 3 of 7

Find the input impedance of the RLC circuit.

$$\mathbf{Z} = \left(\frac{1}{j\omega C} \right) \parallel (R + j\omega L)$$

$$\begin{aligned}
 &= \frac{(R + j\omega L) \left(\frac{1}{j\omega C} \right)}{R + j\omega L + \frac{1}{j\omega C}} \\
 &= \frac{\frac{R + j\omega L}{j\omega C}}{\frac{j\omega RC + (j\omega)^2 LC + 1}{j\omega C}} \\
 &= \frac{R + j\omega L}{(1 - \omega^2 LC) + j\omega RC}
 \end{aligned}$$

Rationalize the expression

$$\begin{aligned}
 \mathbf{Z} &= \frac{R + j\omega L}{(1 - \omega^2 LC) + j\omega RC} \times \frac{(1 - \omega^2 LC) - j\omega RC}{(1 - \omega^2 LC) - j\omega RC} \\
 &= \frac{(R + j\omega L)((1 - \omega^2 LC) - (j\omega RC)(R + j\omega L))}{(1 - \omega^2 LC)^2 + (\omega RC)^2} \\
 &= \frac{R + j\omega L - \omega^2 RLC - j\omega^3 L^2 C - j\omega R^2 C + \omega^2 RLC}{(1 - \omega^2 LC)^2 + (\omega RC)^2} \\
 &= \frac{R - \omega^2 RLC + \omega^2 RLC + j(\omega L - \omega^3 L^2 C - \omega R^2 C)}{(1 - \omega^2 LC)^2 + (\omega RC)^2}
 \end{aligned}$$

Simplified further...

$$\begin{aligned}
 \mathbf{Z} &= \frac{R + j(\omega L - \omega^3 L^2 C - \omega R^2 C)}{(1 - \omega^2 LC)^2 + (\omega RC)^2} \\
 &= \frac{R}{(1 - \omega^2 LC)^2 + (\omega RC)^2} + j \frac{(\omega L - \omega^3 L^2 C - \omega R^2 C)}{(1 - \omega^2 LC)^2 + (\omega RC)^2}
 \end{aligned}$$

At resonance condition a RLC circuit in which capacitive and inductive reactances are equal in magnitude. Thus, results imaginary part is zero.

$$\frac{\omega L - \omega^3 L^2 C - \omega R^2 C}{(1 - \omega^2 LC)^2 + (\omega RC)^2} = 0$$

The value of ω that satisfies this condition is called the resonant frequency ω_0 .

$$\begin{aligned}
 \frac{\omega_0 L - \omega_0^3 L^2 C - \omega_0 R^2 C}{(1 - \omega_0^2 LC)^2 + (\omega_0 RC)^2} &= 0 \\
 \omega_0^3 L^2 C - \omega_0 R^2 C - \omega_0 L &= 0 \\
 \omega_0 (-\omega_0^2 L^2 C - R^2 C + L) & \\
 \omega_0^2 L^2 C + R^2 C - L &= 0 \\
 \omega_0^2 L^2 C &= L - R^2 C \\
 \omega_0^2 &= \frac{L - R^2 C}{L^2 C} \\
 &= \frac{L}{L^2 C} - \frac{R^2 C}{L^2 C} \\
 &= \frac{1}{LC} - \frac{R^2}{L^2}
 \end{aligned}$$

Simplify the equation.

$$\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

Thus, the resonant frequency ω_0 is $\boxed{\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}}$.

(b)

Consider the following circuit:

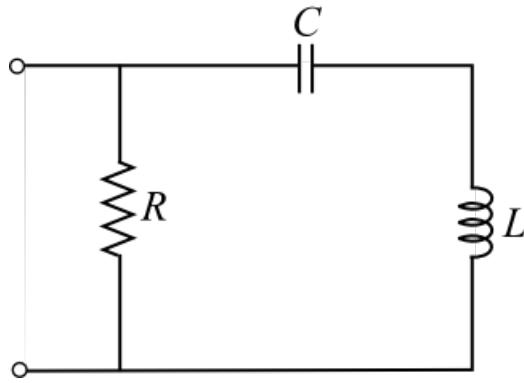


Figure 3

Step 5 of 7

The frequency domain equivalent circuit is as shown below.

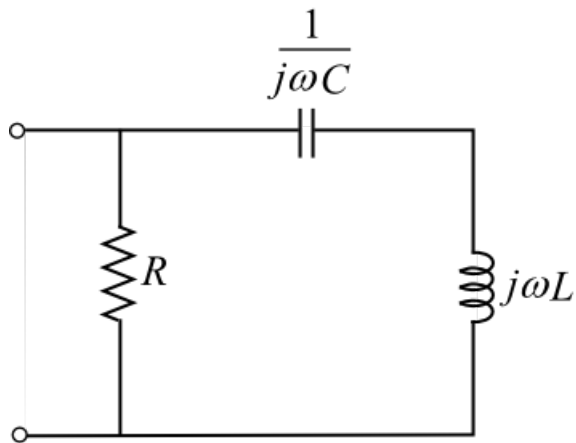


Figure 4

Step 6 of 7

Find the input impedance of the RLC circuit.

$$\begin{aligned} \mathbf{Z} &= (R) \parallel \left(\frac{1}{j\omega C} + j\omega L \right) \\ &= \frac{R \left(\frac{1}{j\omega C} + j\omega L \right)}{R + \frac{1}{j\omega C} + j\omega L} \\ &= \frac{R \left(\frac{1 + (j\omega)^2 LC}{j\omega C} \right)}{\frac{j\omega RC + 1 + (j\omega)^2 LC}{j\omega C}} \\ &= \frac{R(1 - \omega^2 LC)}{(1 - \omega^2 LC) + j\omega RC} \end{aligned}$$

Rationalize the expression

$$\begin{aligned} \mathbf{Z} &= \frac{R(1 - \omega^2 LC)}{(1 - \omega^2 LC) + j\omega RC} \times \frac{(1 - \omega^2 LC) - j\omega RC}{(1 - \omega^2 LC) - j\omega RC} \\ &= \frac{R(1 - \omega^2 LC)^2 - j\omega R^2 C(1 - \omega^2 LC)}{(1 - \omega^2 LC)^2 + (\omega RC)^2} \\ &= \frac{R(1 + (\omega^2 LC)^2 - 2\omega^2 LC) - j\omega R^2 C + j\omega^3 R^2 LC^2}{(1 - \omega^2 LC)^2 + (\omega RC)^2} \\ &= \frac{R - \omega^2 RLC + \omega^2 RLC + j(-\omega R^2 C + \omega^3 R^2 LC^2)}{(1 - \omega^2 LC)^2 + (\omega RC)^2} \\ &= \left(\frac{R - \omega^2 RLC + \omega^2 RLC}{(1 - \omega^2 LC)^2 + (\omega RC)^2} \right) + j \left(\frac{-\omega R^2 C + \omega^3 R^2 LC^2}{(1 - \omega^2 LC)^2 + (\omega RC)^2} \right) \end{aligned}$$

Step 7 of 7

At resonance condition a RLC circuit in which capacitive and inductive reactances are equal in magnitude. Thus, results imaginary part is zero.

$$\frac{-\omega R^2 C + \omega^3 R^2 LC^2}{(1 - \omega^2 LC)^2 + (\omega RC)^2} = 0$$

The value of ω that satisfies this condition is called the resonant frequency ω_0 .

$$\begin{aligned} \frac{-\omega_0 R^2 C + \omega_0^3 R^2 LC^2}{(1 - \omega_0^2 LC)^2 + (\omega_0 RC)^2} &= 0 \\ -\omega_0 R^2 C + \omega_0^3 R^2 LC^2 &= 0 \\ \omega_0 (-R^2 C + \omega_0^2 R^2 LC^2) &= 0 \\ -R^2 C + \omega_0^2 R^2 LC^2 &= 0 \end{aligned}$$

Simplify the equation.

$$\begin{aligned} \omega_0^2 R^2 LC^2 &= R^2 C \\ \omega_0^2 LC &= 1 \\ \omega_0^2 &= \frac{1}{LC} \\ \omega_0 &= \frac{1}{\sqrt{LC}} \end{aligned}$$

Thus, the resonant frequency ω_0 is $\boxed{\frac{1}{\sqrt{LC}}}$.

